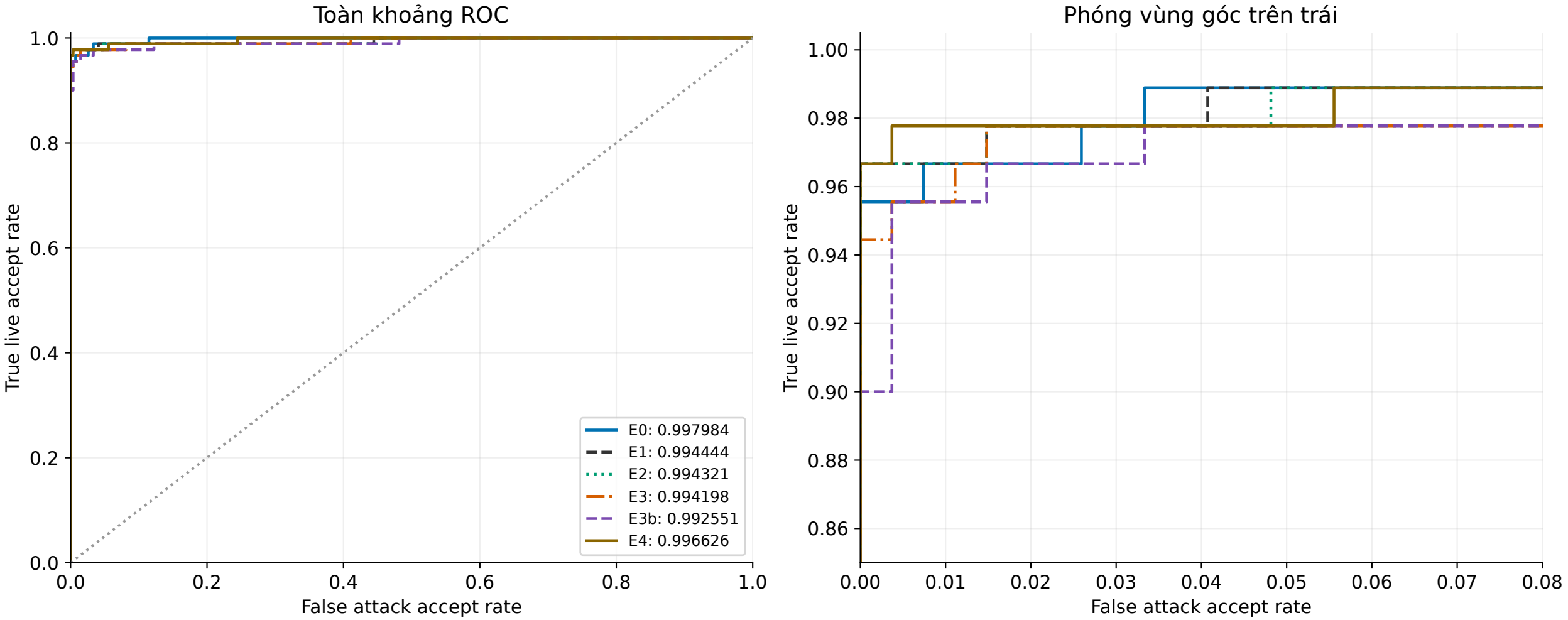

CASIA development protocol · seed 42 · video mean of 20 frames · 270 attack / 90 live test videos. ROC/AUC uses video ranking across thresholds; ACER uses each model's fixed validation threshold. Right panel is an explicitly labelled zoom.